HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed, calibrated, and delivered quantitative models for market microstructure and transaction cost analysis, focusing on robust out-of-sample performance across US, EMEA, and APAC equities.
- Owned the full model lifecycle for quantitative research, including performance monitoring, live-to-historical reconciliation, drift analysis, and recalibration, ensuring sustained accuracy.
- Wrote, tested, and maintained production-quality Python code for quantitative research and live analytics, adhering to strong engineering standards including unit testing and documentation.
- Partnered with portfolio managers and traders to understand business needs and validate model behavior in production environments, leading to actionable insights for trading strategies.
- Developed and supported data pipelines using Python and SQL for processing large financial datasets, ensuring data integrity for model training and live analytics platforms.
- Containerized quantitative research applications using Docker and deployed them to a Kubernetes cluster on AWS, enabling scalable and reproducible research environments.
- Collaborated with team members on the design and architecture of a quantitative research platform, contributing to system architecture and technical direction.

SKILLS

Quantitative Research – Market Microstructure, Factor Risk Modeling, Transaction Cost Analysis, Alpha Signals, Out-of-Sample Performance

Programming & Data Analysis – Python, SQL, Pandas, NumPy, R, RStudio

Machine Learning & Statistics – Gradient Boosted Trees, Deep Learning (PyTorch), Time-Series Models, Statistical Modeling, Hypothesis Testing, Regression

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Linux, Bash

Development Tools & Practices – Git, Jenkins, CI/CD, Unit Testing, Integration Testing, Documentation

Financial Domain – Equity Markets, Trading Efficiency, Execution Quality, Portfolio Management, Live Analytics, kdb+

Data Visualization & BI – Power BI, Tableau, DAX, Power Query

PROJECTS

Equity Market Microstructure Analysis Platform

- Developed Python-based data processing pipelines to ingest and clean tick data for US and EMEA equities, handling approximately 5 million records daily.
- Implemented models for order book imbalance and liquidity analysis using scikit-learn, achieving a mean absolute error of ~0.015 on simulated order flow data.
- Built interactive dashboards in Power BI to visualize microstructure metrics and model outputs for research team review, facilitating faster hypothesis generation.

Transaction Cost Analysis (TCA) Model Refinement

- Engineered Python scripts to extract and aggregate trade execution data from enterprise systems, focusing on identifying sources of slippage and delay.
- Calibrated a gradient-boosted tree model (XGBoost) using historical trade data to predict execution costs for different order types, reaching an R-squared of 0.78 on validation sets.
- Collaborated with quantitative traders to refine model parameters based on live trading feedback and observed market conditions.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)